

First Year

Fall Semester					Spring Semester				
Course Code	Course Title	Credit Hrs			Course Code	Course Title	Credit Hrs		
		Th	Pr	Total			Th	Pr	Total
CS-114	Fundamentals of Computer Engineering	3	1	4	CS-116	Object Oriented Programming	3	1	4
CS-115	Computer Programming	3	1	4	CS-117	Discrete Structures	3	0	3
EE-120	Basic Electrical Engineering	3	1	4	PH-122	Applied Physics	3	1	4
MT-114	Calculus	3	0	3	EL-103	Basic Electronics	3	1	4
HS-111	Functional English	2	0	2	HS-205	Islamic Studies OR	2	0	2
HS-106	Pakistan Studies OR	1	0	1	HS-209	Ethical Behaviour			
HS-107	Pakistan Studies (For Foreigners)								

Second Year

Fall Semester					Spring Semester				
Course Code	Course Title	Credit Hrs			Course Code	Course Title	Credit Hrs		
		Th	Pr	Total			Th	Pr	Total
CS-218	Data Structures & Algorithms	3	1	4	CS-221	Computer Organization & Design	3	1	4
CS-220	Digital Logic Design	3	1	4	CS-215	Signals and Systems	2	1	3
CS-219	Computer Engineering Workshop	0	1	1	CS-222	Database Management Systems	3	1	4
MT-224	Complex Variables & Fourier Analysis	3	0	3	MT-222	Linear Algebra and Ordinary Differential Equations	3	0	3
EE-217	Circuit Theory	2	0	2					
HS-218	Business Communication	2	1	3	HS-219	Professional Ethics	2	0	2

Third Year

Fall Semester					Spring Semester				
Course Code	Course Title	Credit Hrs			Course Code	Course Title	Credit Hrs		
		Th	Pr	Total			Th	Pr	Total
CS-328	Computer Architecture	3	0	3	CS-301	Microprocessor Based System Design	3	1	4
CS-326	Software Engineering	3	1	4	CS-329	Operating Systems	3	1	4
CS-327	Computer Communication Networks	3	1	4	MT-442	Numerical Methods	3	0	3
CS-323	Artificial Intelligence	3	1	4	EF-305	Engineering Economics and Management	3	0	3
MT-335	Probability & Statistics	3	0	3	CS-3XX	Elective-1	3	1	4

Final Year

Fall Semester					Spring Semester				
Course Code	Course Title	Credit Hrs			Course Code	Course Title	Credit Hrs		
		Th	Pr	Total			Th	Pr	Total
CS-419	Digital Signal Processing	3	1	4	CS-438	Computer System Modelling	3	0	3
CS-431	Digital System Design	3	1	4	CS-432	Distributed Computing	2	1	3
CS-425	Entrepreneurship for Computer Engineers	3	0	3	MG-482	Organizational Behaviour	3	0	3
CS-4XX	Elective-2	2	0	2	CS-4XX	Elective-3	2	1	3
CS-406	*Computer Engineering Project	0	3	3	CS-406	Computer Engineering Project	0	3	3
CS-3XX	Elective - 1 (3+1)	CS-4XX	Elective - 2 (2+0)		CS-4XX	Elective - 3 (2+1)			
CS-330	Digital Communication Systems	CS-433	Internet Computing		CS-426	Computer System Security			
CS-324	Machine Learning	CS-434	Bioinformatics		CS-436	Computer Vision			
CS-325	Software Development & Testing	CS-435	Software Project Management		CS-437	Parallel Programming			

* Duration one academic year: Requires literature survey and preliminary work during this Semester. Final exam is conducted in the last semester.

First Year**Fall Semester****CS-114 FUNDAMENTALS OF COMPUTER ENGINEERING (3+1)**

Number Systems; Digital Logic Gates, Classification of Computer Systems; Von Neumann Architecture; Structure and Organization of Computers and Computer Systems; Computer Peripherals; Classification of Software Systems; Introduction to Software Development Process; Levels of Programming Languages; Data Communication and Computer Networking Fundamentals; Introduction to Databases and Information Systems; Introduction to Operating Systems; Pseudo-programming; Recent Trends in the field of Computer Engineering, Augmented Reality, Virtual Reality, IoT, Big Data Computing, Cloud Computing; Society and Information Technology, Ethical issues related to Computers

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CS-115 COMPUTER PROGRAMMING (3+1)

Introduction to Programming Languages; Language Building Blocks: Variables & Constants, Operators & Expressions, Input/processing/output; Basic Data Types; Decision Control Structures; Loop Control Structures; Arrays and Collections; User-Defined and Built-in Functions; Files; Program Debugging and Testing

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EE-120 BASIC ELECTRICAL ENGINEERING (3+1)

Fundamentals of Electric Circuits - Charge, Current, Voltage and Power, Voltage and Current Sources, Ohm's Law; Voltage and Current Laws - Nodes, Paths, Loops and Branches, Kirchhoff's Current Law, Kirchhoff's voltage Law, the Single-Loop Circuits, the Single Node-Pair Circuit, Series and Parallel Connected Independent Sources, Resistors in Series and Parallel, Voltage and Current Division.

Basic Nodal and Mesh Analysis - Multi-Nodal Analysis, the Super node, Mesh Analysis, the Super mesh Circuit Analysis Techniques - Linearity and Superposition, Source Transformations, Thevenin and Norton Equivalent Circuits,

Maximum Power Transfer, Delta- Wye Conversion

Capacitors and Inductors - Capacitor, Inductor, Inductance and Capacitance Combination

Basic RL and RC Circuits - The Source-Free RL Circuit, Properties of the Exponential Response, the Source-Free RC Circuit, the Unit-Step Function, Driven RL Circuits, Natural and Forced Response, Driven RL Circuits

The RLC Circuit - The Source-Free Parallel Circuit, the over damped Parallel RLC Circuit, Critical Damping, the under damped Parallel RLC Circuit, the Source- Free Series RLC Circuit, the Complete Response of the RLC Circuit, the Lossless LC Circuit

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MT-114 CALCULUS**(3+0)**

Set and Functions: Define rational, irrational and real numbers; rounding off a numerical value to specified value to specified number of decimal places or significant figures; solving quadratic, and rational inequalities in involving modulus with graphical representation; Definition of set, set operations, Venn diagrams, DeMorgan's laws, Cartesian product, Relation, Function and their types (Absolute value, greatest integer and combining functions). Graph of some well-known functions. Limit of functions and continuous and discontinuous functions with graphical representation; Differential Calculus: Differentiation and Successive differentiation and its application: Leibnitz theorem; Taylor and Maclaurin theorems with remainders in Cauchy and Lagrange form, power series; Taylor and Maclaurin series, L Hopitals rule, extreme values of a function of one variable using first and second derivative test, asymptotes of a function, curvature and radius of curvature of a curve, partial differentiation, exact differential and its application in computing errors, extreme values of a function of two variables with and without constraints, solution of non-linear equation using Newton Raphson's method; Integral Calculus: Indefinite integrals and their computational techniques, reduction formulae, definite integrals and their convergence; Beta and Gamma functions and their identities, applications of integration; Centre of pressure and depth of centre of pressure. Sequence & Series: Sequence, Infinite Series, Application of convergence tests such as comparison, Root, Ratio, Raabe's and Gauss tests on the behaviour of series; Complex Number: Argand diagram, De Moivre formula, root of polynomial equations, curve and regions in the complex plane, standard functions and their inverses (exponential, circular and Hyperbolic functions).

[\(back\)](#)**HS-111 FUNCTIONAL ENGLISH****(2+0)**

Speaking and Listening: Listening actively through the use of skills and sub skills, and in a variety of situations. Speaking: Fluency and confidence building through group discussions, role plays and public speaking; Vocabulary development: Tips / strategies in vocabulary enhancement, Practice in vocabulary development; Reading: Reading skills, Sub skills, Reading strategies, Reading practice through variety of reading texts and comprehension exercises, Précis writing; Writing: Note taking: Techniques for taking notes from lectures, from books (integrated with listening & reading). Process of Writing with practice in pre writing strategies, in revising, and in, editing for grammar. Writing well- structured and effective paragraphs, essays and letters (routine communication) using proper writing mechanics; Writing descriptions, narrations, cause and effect, compare and contrast etc.

[\(back\)](#)**HS-106 PAKISTAN STUDIES****(1+0)**

Historical and ideological Two Nation Theory: Claim of Muslims of being a separate nation from Hindus, perspective of Pakistan based upon cultural diversity.

Cultural diversity and interests as bases for the Movement: demand of Pakistan – Lahore resolution.

Creation of Pakistan: Factors leading to the creation of Pakistan, Quaid-e-Azam and the demand of Pakistan

Constitutional Process: Constitutional and Political developments in Pakistan 1947-1973. Salient features of the Constitutions 1956, 1962 and 1973 and amendments.

Land of Pakistan: Geo-physical conditions, Geo-political and strategic importance of Pakistan, Natural resource, viz: mineral, water and power.

Contemporary issues in Pakistan: A brief survey of Pakistan Economy: problems, issues and future prospects.

Pakistani Society and Culture-Broad features with emphasis on youth role in the development of Pakistan.

Literacy and education in Pakistan: problems and issues. State of Science and Technology in Pakistan: A comparison with other countries with special reference to the Muslim world.

Environmental issues in Pakistan: government policies and measures and suggestions for improvement.

Pakistan's role in the preservation of nature through international conventions / treaties

Human Rights in Pakistan: Pakistan's response to human rights issues at national & international levels.

Pakistan's Foreign Policy Urbanization in Pakistan - problems and issues

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HS-107 PAKISTAN STUDIES (FOR FOREIGNERS)

(1+0)

Land of Pakistan: Land and People, Strategic Importance, Important and Beautiful Sights: Natural Resources (some portion of Economics of Pakistan)

A Brief Historical Background: A brief historical survey of Muslim Community in the Sub- Continent, British Rule and its Impacts, Indian Reaction, Two-Nation Theory, its Origin and Development, Factors leading towards the Demand of a separate Muslim State, Creation of Pakistan.

Government & Political Development in Pakistan: Constitution of Pakistan, A brief outline, Governmental Structure, Federal and Provincial, Local Government Institutions, Political History and its brief account.

Pakistan & the Muslim World: Relations with Muslim Countries

Language and Culture: Origin of Urdu Language, Influence of Arabic and Persian on Urdu Language and Literature, A short history of Urdu Literature, Dominant Cultural Features

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Spring Semester

CS-116 OBJECT ORIENTED PROGRAMMING

(3+1)

Difference between Conventional and Object Oriented Programming; Data Abstraction; Objects, Classes and Methods; Constructors; Destructors; Operator Overloading; Generic Programming: Class and Function Templates; Inheritance; Multiple Inheritance; Polymorphism; Aggregation; Exception Handling; Program Debugging and Testing, Event Logging

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CS-117 DISCRETE STRUCTURES

(3+0)

Purpose and role of discrete structures in computer engineering, contrasts between discrete-time models and continuous-time models; Sets - Venn diagrams, complements, Cartesian products, power sets, cardinality and countability; Relations - reflexivity, symmetry, transitivity, equivalence relations; Functions - one-to-one, onto, inverses, composition functions; Basic logic - propositional logic, logical connectives, truth tables. Predicate logic, universal and existential quantification, application of logic to computer engineering; Proof techniques - structures of formal proofs, direct proofs, proof by counterexample, contraposition and contradiction; Mathematical induction and strong induction; Introduction to group theory - groups and semi-groups; Basics of counting - permutations and combinations, counting arguments rule of products, rules of sums, discrete probability, generating functions, recurrence relation.

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PH-122 APPLIED PHYSICS

(3+1)

Introduction: Types of Errors and Error Calculation, Graphical Techniques (Log, semi-log and other non-linear graphs).

Vectors: Coordinate Systems, Review of vectors, Vector Differentiation (Ordinary and Partial Differentiation), Vector Integrations.

Mechanics: Motion under Constant Acceleration, Newton Laws and their Applications, Frictional Forces, Work-Energy Theorem, Law of Conservation of Mechanical Energy, Angular Momentum.

Electrostatics and Magnetism: Coulombs Law, Continuous charge distribution, Electrostatic potential energy of discrete charges, Gauss's Law, Electric field around conductors, Magnetic fields, Magnetic force on current, Hall effect, Biot-Savart Law, Ampere's Law, Field of rings and coils, Magnetic dipole, Diamagnetism, Paramagnetism and Ferromagnetism. Semiconductor Physics: Energy levels in a semiconductor, Hole concept, Intrinsic and Extrinsic regions, Law of Mass Action, p-n junction, Transistor. Waves and Oscillations: Simple Harmonic Oscillator, Damped Harmonic Oscillation, Forced Oscillation and Resonance, Type of Waves and Superposition Principle, Wave Speed on a stretched string. Optics and Lasers: Huygen's Principle, Two-slit interference, Single-Slit Diffraction, Resolving power of Optical Instruments Principles for laser action, Types of laser, Applications of laser. Modern Physics: Planck's explanations of Black Body Radiation Photoelectric Effect, Compton Effect, Bohr's

Theory of Hydrogen Atom, Atomic Spectra, Reduced Mass, De-Broglie Hypothesis, Electron Microscope, Atomic Nucleus and Properties of Nucleus, Radioactive Decay and Radioactive Dating, Radiation Detection Instruments, Nuclear Reactions and Nuclear Reactor.

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EL-103 BASIC ELECTRONICS

(3+1)

PN Junction Diode: Introduction. PN junction diode, Unbiased diode, Barrier potential, Diffusion & drift current, Forward & reverse bias, Minority carrier current. Diode models: Ideal, practical & complete, Diode characteristics, Load line, Diode current equation; Capacitive effect on diode operation at high-frequency, Transient current, Temperature effect on diode operation; Diode applications; Rectifier, Switch, Communication, Wave-shaping, Voltage multiplier etc; Breakdown diode, Voltage regulator, Power Supply.

FET: Field effect transistor; Device, Structure & Physical operation of the MOSFET, Current-voltage characteristics; DC analysis of MOSFET circuits, MOSFET as an amplifier and as a switch, biasing in MOS amplifier circuits.

Operational Amplifiers: Terminal characteristics only, ideal op-amp characteristics, inverting and non-inverting configurations, op-amp applications like weighted summer, difference amplifier, instrumentation amplifier, differentiator, integrator, logarithmic amplifier etc; Non ideal characteristics like slewing, DC input offset voltage, input biasing current etc.

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HS-205 ISLAMIC STUDIES

(2+0)

Quranic Verses - Tauheed, (Al-Ambiyah-22, Al-Baqarah-163 and 164). Prophethood (Al-Imran- 79, Al-Huda-7, Al-Maidah-3). Here-After (Al-Baqarah-48 and one Hadith). Basic Islamic Practices (Al-Mu'minin-1 and 111, and two Ahadith). Amer-Bil-Ma'Roof Wa-Nahi Anil Munkar. The Concept of Good and Evil; Importance and Necessity of Da'wat-e-Deen (Al- Imran-110). Method of Da'wat-e-Deen (An-Nehl-125, Al-Imran-64, Al-An'an-108, and two Ahadith). Kasb-e-Halal (Taha-81, Al-A'raf-32 and 33, Al-Baqarah-188 and two Ahadith). Huquq-ul-Ibad: Protection of Life (Al-Madinah-32), Right to Property (Al-Nisa-29), Right to Respect and Dignity (Al-Hujrat-11 and 12), Freedom of Expression (Al-Baqarah-256), Equality (Al-Hujarat-13), Economic Security (Al-Ma'ar ij-24 and 25); Employment Opportunity on Merit (An-Nisa-58); Access to Justice (An-Nisa-135); Women's Rights (An-Nehl-97, Al-Ahzab-35, An-Nisa-07); Relation with Non-Muslims (Al-Muhammad-8 and 9, Al-Anfa'al-61 and The Last Sermon of Hajj of Holy Prophet (SAW); Relevant Extracts. Seerat (life) of the Holy Prophet (SAW) - Birth, Life at Makkah, Declaration of Prophet hood, Preaching and its Difficulties, Migration to Madinah, Brotherhood (Mawakhat) and Madinah Charter, the Holy Wars of the Prophet (SAW) (Ghazwat-e-Nabawi), Hujrat-ul-Wida, the Last Sermon of Khutbatulwida (Translation and Important Points). Islamic Civilization - (a) In the Sub Continent: Pre-Islamic Civilizations, the Political , Social and Moral Impact of Islamic Civilizations. (b) In the World: Academic, Intellectual, Social and Cultural Impact of Islamic Civilization.

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HS-209**ETHICAL BEHAVIOUR****(2+0)**

Nature, Scope and Methods of Ethics; Ethics and Religion; Ethical Teachings of World Religions; Basic Moral Concepts, Rights and Wrong, Good and Evil; An Outline of Ethical Systems in Philosophy; Hedonism, Utilitarianism, Rationalism and Kant; Self Realization Theories, Intuitionism; Islamic Moral Theory: Ethics of Quran and its Philosophical Basis; Ethical, Precepts from Quran and Hadith and Promotion of Moral Values in Society.

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Second Year

Fall Semester

CS-218 DATA STRUCTURES & ALGORITHMS

(3+1)

Basic Concepts - Introduction and classification of Data Structures; Basic operations, Classification of Algorithms - Classification by implementation; Classification by design paradigm; Basics of Complexity Analysis - Rate of Growth of Complexity of Algorithms; Asymptotic notations; Time-Space Trade Offs. String Processing - Operations on strings; Word Processing; Pattern Matching Algorithms. Arrays - One-dimensional Arrays: Searching and Sorting Algorithms; Multi-dimensional Arrays: Matrix Multiplication, Sparse Matrices. Stacks, Queues and Recursion - Basic concepts and functions; Polish Notation; Quick-sort; Deques; Priority Queues; Factorial Calculation; Fibonacci Series; Ackermann Function, Towers of Hanoi. Linked Lists - Definition and Representation; Traversal and Searching; Insertion; Deletion; Circular Lists; Doubly Linked Lists. Trees - Terminology; Representation in memory; Binary Trees: Traversal Algorithms; Binary Search Trees, Heaps; Heap-sort algorithm. Graphs - Terminology; Representation in memory; Traversal Algorithms; Shortest Path Algorithms; Sorting and Searching - Sorting Algorithms; Hashing

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CS-220 DIGITAL LOGIC DESIGN

(3+1)

Introduction to Digital Design: Basic Logic Gates, Logic Families; Boolean Algebra; Gate-Level Minimization of Boolean Functions using Karnaugh Map; NAND and NOR Implementation, Duality, Active Low Logic; Combinational Logic Circuit Design: Adders, Decoders, Encoders, Multiplexers, DeMultiplexers, Magnitude Comparators and Binary Code Converters; Storage Devices and Sequential Logic: Latches and Flip Flops, Shift Registers, Asynchronous Counters, Mealy and Moore Machines, Synthesis of Sequential Logic Circuits, Synchronous Counters, State Minimization techniques (pair chart) for completely specified sequential circuits; Memories and Programmable Logic Devices: Read Only Memory, Programmable Array Logic and Programmable Logic Array

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CS-219 COMPUTER ENGINEERING WORKSHOP

(0+1)

Android development and environment, Introduction to Java programming, Building basic Android App, Basic Android User Interface (UI), Advanced Android User Interface, Multimedia in Android, Using SQL databases in Android, Data Storage, Retrieval and Sharing, Mapping and Location based Services, Using background Worker Threads

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MT-224 COMPLEX VARIABLE & FOURIER ANALYSIS**(3+0)**

Infinite Series: Application of convergence tests such as comparison, Root, Ratio, Raabe's and Gauss tests on the behaviour of series.

Complex Variable: Limit, continuity, zeros and poles of a complex function. Cauchy-Reimann equations, conformal transformation, contour integration.

Laplace Integral & Transformation: Definition, Laplace transforms of some elementary functions, first translation or shifting theorem, second translation or shifting theorem, change of scale property, Laplace transform of the n th order derivative, initial and final value theorem Laplace transform of integrals, Laplace transform of functions $t^n F(t)$ and $F(t)/t$, Laplace transform of periodic function, evaluation of integrals, definition of inverse Laplace transform and inverse transforms, convolution theorem, solutions of ordinary differential and partial differential equations using Laplace transform (I.V.P's & B.V.P's), Z and Inverse Z –transformations, properties of Z - transformation and applications

Fourier series: Introduction to Fourier series. Euler Fourier formulae, even and odd functions, application of Fourier series, Fourier transform and fast Fourier transform and properties with applications.

[\(back\)](#)**EE-217 CIRCUIT THEORY****(2+0)**

Elementary Transient Analysis - Differential and integral forms of circuit equations, Initial voltage on a capacitor, Initial current in an inductor, First -order circuits, Solution of single first order differential equations, particular and total solution of second order linear time invariant differential equations. Matrix Analysis - Systematic formulation of network equations, Loop variable analysis, State variable analysis, formulation of state equations, source transformations Duality. Elementary Time Functions - Introduction to singularity functions, The impulse functions and response, The unit step function and response, Ramp function, Exponential function & response. Exponential Excitation and the Transformed Network - Representation of excitations by exponentials functions, Single element response, Forced response with exponential excitation, Introduction to the transformed network, Driving point impedance and admittance. Two Port Network - Introduction, Characterization of linear time invariant two-ports by six sets of parameters, Relationship among parameter sets. Networks Functions and Frequency Response - The concept of complex frequency, transform impedance and transform circuits, Network functions. Poles and zeros of network functions, Restrictions on pole and zero transfer function, magnitude and phase.

[\(back\)](#)**HS-218 BUSINESS COMMUNICATION**

Foundations of Business Communication: Definitions: communication, organization, business; understanding the need and scope of business, professional and organizational communication, Conditions, properties, process, tools, modes, levels, types of communication; Principles of Effective Communication & Building goodwill (You-attitude, positive emphasis and unbiased language); Listening, non-verbal communication. Communication dilemmas and problems; Feedback and its types; Audience Analysis Oral Communication: Group Discussions and

interpersonal skills, Meetings, Interviews, Making presentations Business & Technical Writing: Types of messages: Formats (Letter and memorandum); Letter and memorandum elements and formats. Three Types of Business Messages (routine, negative and persuasive communications). Organizational Plans: Direct, Indirect & AIDA approach; Writing business messages (e-mails, inquiries, requests, replies, regrets, declining offers, letters, routine messages, etc.); Meetings: notice, agenda and minutes. Job applications and resumes. Research / scientific reports (structure, layout, writing process)

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Spring Semester**CS-221 COMPUTER ORGANIZATION & DESIGN (3+1)**

Introduction & Motivation: Instruction Set Architecture (ISA) as an interface between hardware and software; ISA: Instruction Formats, Addressing Modes, Translation of High Level Language Program into Machine Language; Classes and Comparative Study of ISAs: Accumulator-based, Stack-based, Register-Memory & Register-Register Architectures; Instruction Encoding Techniques; Measuring and Reporting Computer Performance: Common Pitfalls, Amdahl's Law; Computer Arithmetic: Integer and Floating-Point; Basic Processor Design: Single-Cycle and datapath & control; Memory System Design: Memory Hierarchy, Cache Memory; Input/Output: Bus Standards, Arbitration Schemes, Programmed I/O; Interrupt-Driven I/O. Interrupt Processing, Direct Memory Access (DMA); Emerging Architectures.

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CS-215 SIGNALS AND SYSTEMS (2+1)

Fundamental concepts: Signals and Systems, Continuous-Time Signals, Discrete-Time Signals, Examples of Systems, Basic System Properties; Systems defined by Differential Equations: Linear Input / Output Differential Equations with Constant Coefficients, System Modelling, Systems defined by nonlinear equations; Convolution Representation of Linear Time-Invariant Continuous-Time Systems: Convolution of Continuous-Time Signals, Linear Time-Varying Systems; Fourier Analysis: Representation of Signals in Terms of Frequency Components, Fourier Series Representation of Periodic Signals, Fourier Transform, Properties of Fourier Transform; Frequency-Domain Analysis of Systems: Response to a Sinusoidal Input, Response to Periodic Inputs, Response to Aperiodic Inputs, Analysis of Filters; Laplace Transform: Properties, Computation, Transfer Function Representation and Block Diagrams, Stability,

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CS-222 Database Management Systems (3+1)

Database basics: Benefits of Database; Data Models; Data Modelling and Database Design; Database System Architectures; Relational Databases: Entity-Relationship Model; Relational Algebra; Relational design principles; Normalization; Relational Database Implementation – SQL; Database Development Process: Indexing and Hashing; Transaction Management; Recovery and Concurrency Control; Security and Auditing, Introduction of No-SQL and Non-Relational Databases; design and implementation philosophy of No-SQL Databases; CAP theorem; Introduction to Distributed Databases, Decision Support Systems, Data Warehousing and Data Mining

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MT-222 Linear Algebra & Ordinary Differential Equations**(3+0)**

Linear Algebra: Linearity and linear dependence of vectors, basis, dimension of a vector space, field matrix and type of matrices (singular, non-singular, symmetric, non-symmetric, upper, lower, diagonal), Rank of a matrix using row operations and special method, echelon and reduced echelon forms of a matrix, determination of consistency of a system of linear equation using rank, matrix of linear transformations, Eigen value and Eigen vectors of a matrix, Diagonalization, Applications of linear algebra (Scaling, translation, rotation and projection) with graphical representation.

Introduction to Ordinary Differential Equations (ODE): The Concepts & Terminologies: Order and Degree; Linearity & Non-linearity; A Brief Classification of ODEs; Formulation of ODEs: Concrete Examples; Solutions: General & Particular: Concrete Examples & Applications: Initial Value Problems (IYP) and Boundary Value problems (BVP): A Brief Introduction to Issues related to Existence & Uniqueness of Solutions.

The First Order ODEs: Linear and Non-Linear: Variable Separable Cases & Applications: Growth & Decay Problems, Newton's Law of Cooling, Torricelli's Law, Simple Kinematical Dynamical Applications; Exact and Non-Exact ODEs: Solution Procedures and Integrating Factors; the Standard Linear Differential Equation of First Order: Solution Procedures and Applications to RL-Circuits and RC-Circuits, Bernoulli's Equations & Logistical Growth Models. Direction Fields and Euler's and Picard's Iterative Schemes for the 1st Order ODEs,

The Linear Second Order ODEs: Homogeneous and Non-Homogeneous Cases, Linear Second Order Homogeneous ODE with Constant Coefficients: Solution Procedures and the Principle of Linear Superposition and Applications, Mechanical Systems & Electrical Systems, Undamped and Damped Harmonic Oscillators: Linear Second Order Non-Homogeneous ODEs with Constant Coefficients: Solution Procedures and the Principle of General Linear Superposition: Complementary Functions & Particular Solutions, the method of Undetermined Coefficients & Variation of Parameters, Applications: Spring-Mass Systems, Damped & Undamped Harmonic Oscillators with Forcing Terms and their ODEs and Solutions; RCL-Circuits and their ODEs and Solutions; The Physics and Mathematics of the Phenomenon of Resonance in Mechanical & Electrical Systems; Cauchy-Euler ODEs and their Solution Procedures.

Partial Differential Equation: Formation of partial differential equations, Solutions of first order linear and special types of second and higher order differential equations, Homogeneous partial differential equations of order one, Lagrange's multiplier.

Advance Calculus & Vector Calculus: Double & triple integral with application (Area, centroid, moment of inertia) vector differentiation & vector integral with applications, Green & Stokes theorem with applications.

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HS-219 PROFESSIONAL ETHICS**(2+0)**

Introduction to Professional & Engineering Ethics: Definitions - Ethics, Professional Ethics, Engineering Ethics, Business Ethics; Ethics & Professionalism. Need and scope of Engineering and Professional Ethics through case studies; Development of Engineering Ethics & Major issues in Engineering & Professional Ethics; Moral Reasoning & Ethical Frameworks: Ethical Dilemma; Resolving Ethical dilemmas and making Moral Choices; Codes of Ethics (of local and international professional bodies). Moral Theories: Utilitarianism, Rights Ethics and Duty Ethics,

Virtue Ethics Self-Realization & Self Interest; Ethical Problem Solving Techniques: Line drawing, flow Charting, Conflict Problems; case studies and applications; Contemporary Professional Ethics: Professional Responsibilities; Risk and Safety as an Ethical Concern for Engineers, Workplace Responsibilities and Ethics: Teamwork, confidentiality and conflicts of interest, Whistleblowing, Bribe and gift, risk and cost - benefit analyses, gender discrimination and sexual harassment; Environmental Ethics; Computer Ethics & the Internet; Honesty: Truthfulness, trustworthiness, academic and research integrity

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Third Year

Fall Semester

CS-328 COMPUTER ARCHITECTURE (3+0)

Multi-cycle Processor Design, Control unit Design, FSM based implementation of Control Unit, Microprogrammed Control Unit, Instruction Level Parallelism (ILP): Concepts and Challenges - Pipeline Hazards, Avoiding Structural Hazards, Overcoming Data Hazards, Forwarding Unit design, Hazard Detection Unit Design, Basic Compiler Techniques to expose ILP, Branch Prediction, Multiple Issue Architectures - VLIW and Superscalars, Limitations of ILP, Data Level Parallelism in SIMD and Graphic Processing Units (GPUs) Architecture, Detecting and Enhancing Loop-Level Parallelism; Advanced Topics in Memory System Design, Virtual Memory, RAID system

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CS-326 SOFTWARE ENGINEERING (3+1)

Basic definitions; Software attributes; Software process models: waterfall, incremental, reuse-oriented etc.; Software requirements: functional, non-functional, domain; Requirements Engineering Process; Analysis Modelling: Data, Object, Behavioural; Software Project Management: planning and scheduling, risk management and its phases, RMMM; Software Design: general model, architectural design, design patterns; Agile Software Development: extreme programming, agile project management (scrum) Verification & Validation: debugging, software inspection, software testing, types of testing; Software Quality Assurance: Basic approaches, software measurements and metrics, measurement process; Software Cost Estimation: size-oriented metrics function-oriented metrics; algorithmic-cost modelling (COCOMO model). Software Maintenance: Types of Software Maintenance, Software Maintenance Process Models.

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CS-327 COMPUTER COMMUNICATION NETWORKS (3+1)

Introduction to Computer Networks; OSI reference model, the TCP/IP reference model; Packet Switching and Architectures; Circuit Switching and Architectures; Data Link Layer and issues; Error Correction and Congestion Control in networks; Network Layer and Issues (Protocols and Services); IPv4 and IPv6, IP addressing and subnetting; Network Routing; Introduction to Multi-Protocol Label Switching (MPLS); Wireless Networks; Transport Layer and Issues (TCP and UDP); Software Defined Networking (SDN); Virtual Network Functions (VNF). Multimedia networking and streaming services

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CS-323 ARTIFICIAL INTELLIGENCE**(3+1)**

Introduction to AI, Computational Intelligence, Searching Methodologies, First-Order-Logic, Evolutionary Computing: Genetic Algorithms, Evolutionary Strategies, Biology-Inspired Models; Knowledge Representation: Semantic Networks, Frames and Scripts; Reasoning with Imperfect Knowledge, Rule-Based Systems: Modeling, Reasoning Strategies, Conflict Resolution and Rule Matching; Artificial Neural Networks; Vague Notions in Knowledge-Based Systems: Models based on Fuzzy Set Theory; Game Theory, Prospects of Artificial Intelligence.

[\(back\)](#)**MT-335 PROBABILITY AND STATISTICS****(3+0)**

Introduction to Statistics: Introduction, Types of Data & Variables, Presentation of Data, Object, Classifications, Tabulation, Frequency Distribution and their types, Graphical Representation, Simple, Multiple and Component bar diagrams, Pie-chart, Histogram, Frequency polygon and Frequency curves.

Measures of Central Tendency and Dispersion: Statistical Averages, Median, Mode, Quartiles, Range, Moments, Skewness & Kurtosis, Quartile Deviation, Mean Deviation, Standard Deviation, Variance & its coefficient.

Probability: Basic concepts, Permutation & Combination, Definitions of Probability, Laws of Probability, Conditional Probability, Bayes' Rule.

Random Variables and Random Processes: Introduction, Discrete & Continuous Random Variables, Random Sequences and Transformations, Probability Distribution, Probability Density Function, Distribution Function, Mathematical Expectations, Moment Generating Function(M.G.F), Introduction to Random Processes and Time Series, Statistical Averages of Random Processes, Stationary, Auto-Correlation of Wide Sense Stationary Random Processes, Time Averaging, Ergodicity, Markov Chain and Queuing Theory.

Probability Distributions: Introduction, Discrete Probability Distributions, Binomial, Poisson, Hypergeometric & Negative Binomial Distributions, Continuous Probability Distribution, Uniform, Exponential & Normal Distributions.

Sampling and Sampling Distributions: Introduction, Population, Parameter & Statistic, Objects of Sampling, Sampling Distribution of Mean, Standard Errors, Sampling & Non-Sampling Errors, Random Sampling with & without Replacement, Sequential Sampling, Central Limit Theorem.

Statistical Inference and Testing of Hypothesis: Introduction, Estimation, Types of Estimation, Confidence Interval, Tests of Hypothesis, Chi-Square Distribution/Test, One and Two Tails Test.

Simple Regression & Correlation: Introduction, Scatter Diagrams, Correlation & its Coefficient, Regression Lines, Rank Correlation & its Coefficient, Probable Error (P.E).

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Spring Semester

CS-301 MICROPROCESSOR BASED SYSTEM DESIGN (3+1)

Programmer Visible Architecture; Internal Bus Architecture; Pin Functions; Memory Addressing Schemes; Bus Buffering; Bus Cycles; Timing and Sequencing; The Wait State; Clock Generation Circuit; Reset Circuit; Memory Interfacing; Basic I/O Interface; Hardware Single Stepping; Memory Speed Requirements; Programmable Peripheral Interface; Programmable Interval Timer; Hardware Interrupts; Programmable Interrupt Controller; DMA Operations, Microprocessor System Design. Instruction Set Architecture & Addressing Modes; Data Transfer, Arithmetic & Logic; Branch & String Instructions; Program Assembly and Testing; Assembler Directives; Macros; Procedures; Instruction Encoding; Two Pass Assemblers; Software Interrupts

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CS-329 OPERATING SYSTEMS (3+1)

Operating system - objectives and functions; Kernel types and functionality; Processes - state, description, and control; Threads; Operating system design issues; Concurrency - principles; mutual exclusion (hardware support, operating system support); deadlock; Scheduling algorithms; Memory management; secondary storage management; file management - organization and access; I/O management; disk scheduling; Operating system support for multicore architectures

[\(back\)](#)

MT-442 NUMERICAL METHODS (3+0)

Error Analysis: Types of errors (relative, Absolute, inherent, round off, truncation), significant digits and numerical instability, flow chart. Use any Computational tools to Analysis the Numerical Solutions. Solution of Non-linear Equations: Numerical methods for finding the roots of transcendental and polynomial equations (Secant, Newton - Raphson Chebyshev and Graeffe's root squaring methods), rate of convergence and stability of an iterative method. Solution of Linear Equations: Numerical methods for finding the solutions of system of linear equations (Gauss- Elimination, Gauss-Jordan Elimination, triangularization, Cholesky, Jacobi and Gauss - Seidel). Numerical Integration & Differentiation: Computation of integrals using simple Trapezoidal rule, Simpson's rule, Simpson's rule, Composite Simpson's and Trapezoidal rules, computation of solutions of differential equations using(Euler method, Euler modified method, Runge Kutta method of order 4), Numerical Solutions of Partial differential Equations, Optimization problem (Simplex Method), Steepest Ascent and Steepest Descent Methods. Interpolation & Curve Fitting: Lagrange's, Newton, Hermit, Spline, least squares approximation, (Linear and non-linear curves). Linear Operators: Functions of operators, difference operators and the derivative operators, identities. Difference Equations: Linear homogeneous and non- homogeneous difference equations.

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EF 305 ENGINEERING ECONOMICS & MANAGEMENT**(3+0)**

Introduction: Basic Concepts and principles of Economics, Micro- and Macro-economic theory, the problem of scarcity. Basic concepts of Engineering Economy, Financial effectiveness and non-monetary factors, Economic Environment: Consumers and producer goods, Goods and services, Demand & Supply concept. Market Equilibrium, Elasticity of demand, Elasticity of Supply, Measures of Economics worth, Pricesupply-demand-relationship, Revenue, Cost and profit function. Elementary Financial Analysis: Basic accounting equation. Development and interpretation of financial statements-Income Statement, Balance Sheet and Cash Flow, Working capital management, Financial Ratio Analysis. Time Value of Money and Financial Returns: Concepts of simple, compound and effective interest rates, less often than compounding period and more once a year; Present Value, Future Value and Annuities concepts, Uniform gradient and geometric sequence of cash flow; Depreciation and Taxes: Depreciation concept, Economic life, Methods of depreciation, Gain (loss) on the disposal of an asset, Depreciation as a tax shield. Basic cost concepts and Break Even Analysis: Types of costs and cost curves; Determination of Cost/Revenues; Numerical and graphical presentations; Practical applications, BEA as a management tools for achieving financial/operational efficiency Linear Programming: Mathematical statement of linear programming problems, Graphical solutions, Simplex method, Duality Problems. Business Organizations and financial Institutions: Type of ownership, single ownership, partnerships, corporation, type of stocks and joint stock companies, Banking and specialized credit institutions. Project Management: Integration of Organization Strategy with Projects, Defining the project, developing a network plan, managing risk, reducing project time, project selection and comparing alternatives techniques scheduling resources: Introduction to Projection Management and Production Concepts: Basic production function, stages of production, returns to scales, Production lead time, Production rate, capacity, operations, planning and control, order processing, Scheduling, Material requisitions planning, line of balance

[\(back\)](#)**CS-3XX****ELECTIVE-1****(3+1)****CS-330 DIGITAL COMMUNICATION SYSTEMS**

Elements of Modern Digital Communications: Channels, Fundamentals Limitations, Electromagnetic spectrum Signal Analysis: Classification, representation of signals, Baseband systems, filtering, random signal analysis, Analog communication systems: Pulse code modulation, time division multiplexing, Satellite communication, Digital communication: Baseband digital transmission, Digital modulation techniques; Digital encoding; Broadband communication; Error-control and Coding: Error correction and detection techniques, Hamming code, Linear Block Encoding, Information Theory: Entropy, Discrete channels;

[\(back\)](#)**CS-324 MACHINE LEARNING**

The Learning Problem: Components of Learning, Types of Learning; Learning Feasibility, Linear Models: Linear Classification and Regression, Logistic Regression, Non-Linear Transformation; Error and Noise: Error Measures

and Noisy Targets; Training vs. Testing, Theory of Generalization, The VC(Vapnik–Chervonenkis) Dimension: Definition, VC Dimension of Perceptrons, Interpreting VC Dimension, Generalization Bounds; Bias-Variance Tradeoff, Neural Networks: Stochastic Gradient Descent, Backpropagation Algorithm; Overfitting, Regularization, Validation: Model Selection and Cross Validation; Support Vector Machines, Kernel Methods; Introduction to Deep Learning.

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CS-325 SOFTWARE DEVELOPMENT & TESTING

Advanced software process models: formal methods, unified process model; Agile software development: extreme programming, agile process models, scaling agile methods; Embedded software development: embedded systems design, architectural patterns, timing analysis; Software review techniques: defect amplification and removal, review metrics, formal and informal reviews; Software Quality Assurance: SQA elements, tasks, goals and metrics; Statistical SQA; Software Reliability; Testing for conventional applications: white-box testing, black-box testing, model-based testing; Testing for object-oriented applications: Object oriented testing strategies and methods, testing methods applicable at class-level.

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Final Year**Fall Semester****CS-419 DIGITAL SIGNAL PROCESSING (3+1)**

Introduction to Discrete-Time Signals and Systems and an Overview of Practical Applications of Digital Signal Processing; Sampling Theorem; A Review of Discrete-Time Signals and Linear Systems; Transforms (Z, DFT, FFT) and their Applications; Digital Filter Design; Design of FIR and IIR Filters; Implementation Issues; Spectral Analysis.

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CS-431 DIGITAL SYSTEM DESIGN (3+1)

Review of Combinational and Sequential Logic design, Design flow, ASIC based design, FPGA based design, High-level Digital design methodology using HDLs, Behavioural Modelling, Synthesis of Combinational and Sequential Logic, Design and Synthesis of Datapath and Control Logic, Storage devices, Design Optimization techniques in terms of Speed, Area & Power, Post-synthesis Design Validation, Post-synthesis Timing Verification, Fault Simulation and Testing, Design for Testability, Boundary Scan and JTAG Ports, Fault Tolerant Design.

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CS-425 ENTREPRENEURSHIP FOR COMPUTER ENGINEERS (3+0)

The nature and importance of entrepreneurship; The Entrepreneurship and Intrapreneurial mind; The Individual Entrepreneur; Entrepreneurship for IT and E-Business: Role of websites and E-commerce in the development of Global Startups, E-entrepreneurship; Creativity and Business Ideas; Legal Issues for Entrepreneurship; The Business, marketing, organizational and financial plan; Sources of Capital; Entrepreneurial Strategy

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CS-4XX ELECTIVE-2 (2+0)**CS-433 INTERNET COMPUTING**

Introduction to internet systems: History of internet, important organizations and their contributions, standards and protocols, architecture of modern Internet. Review of Internet protocol (v4 and v6); Routing on the Internet: Basics of exterior routing, NAT, MPLS, mobility support in IPv6 and IPv4, multicast transmission through IGMP. Transport layer on the Internet: TCP and UDP functioning and header formats; Programming for the Internet: Internet applications architecture, socket programming and application development using TCP and UDP. Protocols and Services: ICMP, DNS, FTP, HTTP, SMTP, POP3, MIME, Voice over IP; Network management: Importance of auditing, SNMP. Network security: Motives and types of attacks, threats and vulnerabilities, device security

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CS-434 BIOINFORMATICS

Motivation, A Brief History of Bioinformatics, The Biological Foundations of Bioinformatics: The Storage of Genetic Information, Understanding of DNA, RNA & Proteins; Biological Databases: Primary Databases, Nucleotide Sequence Databases, Protein Sequence Databases, Secondary Databases Genotype-Phenotype Databases; Sequence Comparison: Pairwise & Multiple Sequence Comparisons, Algorithms for Database Searching; Probabilistic Models: Bayesian Model, E-Value and Null Model; The Functional Analysis of Genomes.

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CS-435 SOFTWARE PROJECT MANAGEMENT

Introductory concepts; Project planning: pricing, planning process and project scheduling; Risk management process; Managing people; Teamwork; Agile planning; Monitoring and control; Estimation techniques: decomposition techniques, empirical estimation models; Software process improvement: the Capability Maturity Model Integration; Software configuration management: change and version management.

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Spring Semester

CS-438 COMPUTER SYSTEM MODELLING (3+0)

Motivation for Computer Systems Modeling, Performance Evaluation of Computer Systems and Networks, Application of Performance Evaluation; Measurement Techniques; Fundamentals of Queuing Models - Structure and Performance Parameters; Operational Analysis of Queuing Models; General Features of Queuing Models; Birth and Death Processes; M/M/1 and M/G/1 Systems. Dependability Modelling - Analysis of Reliable, Available and High Assurance Systems; Fault Tolerant Techniques; Software Reliability Modelling; Reliability and Availability Modelling; Combinational Modelling; Fault Tolerant Design Techniques; Markov Reliability Models. Petri Net Based Performance Modelling - Classical Petri Nets; Workload Modeling, Simulation Techniques.

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CS-432 DISTRIBUTED COMPUTING (2+1)

Introduction to Distributed Computing; Major trends and challenges; Architectural models for distributed computing; Types of Communication; Remote procedure call; message oriented communication; stream oriented communication; multicast communication; naming in distributed computing; synchronization in distributed computing; Clock synchronization: physical and logical clocks; Mutual exclusion among distributed nodes; Fault tolerance; Distributed file systems; Distributed web based systems; Distributed coordination based systems; Cloud computing; Actor based systems;

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MG-482 ORGANIZATIONAL BEHAVIOUR (3+0)

Foundations of Organizational Behaviour (OB): Management functions, roles, and skills; Effective versus successful managerial activities; replacing intuition with systematic study, Exploring OB challenges and opportunities facing globalization: Improving quality and productivity; improving people skills; managing work force diversity; responding to globalization; empowering people; stimulating innovation and change; coping with temporariness; handling declining employee loyalty; improving ethical behaviour.

Foundations of Individual Behaviour: Individuals & Organizations: Biographical traits and ability; and personality, Perceptions and individual decision making: Understanding perception and its significance, factors influencing perception; linking perception and individual decision making; optimizing decision making model; alternative decision making models; issues in decision making, Values, attitudes and job satisfaction: Importance, sources, types of values; sources and types of attitude; attitude and consistency; measuring job satisfaction; determinants of job satisfaction; effect of job satisfaction on employee performance; ways employees can express dissatisfaction, Motivation- basic concepts and applications

Foundations of Group Behaviour: Group in OB: Defining and classifying groups; stages of group development, work group behaviour; dynamics of groups, Understanding work teams: Team versus group; types of teams, creating high performance teams; turning individuals into team players, Communication: communicating at interpersonal and organizational level, Leadership: basic approaches and contemporary issues, Conflict & negotiation: defining conflict; transition in conflict thought; conflict process; negotiation - strategies, process and

issues.

Foundations of Organizational Structure: Organizational structure and design; work design; work stress, Organizational culture: definition; culture's functions, employees and organizational culture, Organization change and development: forces for change; managing planned change, resistance to change; approaches to managing organizational change.

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CS-4XX

ELECTIVE-3

(2+1)

CS-426

COMPUTER SYSTEMS SECURITY

Cryptographic foundations; Access Control Lists (ACLs); Operating System Security - Principles; Computer systems' structure and its impact on security; Principles and practices for secure system design; Security models: Properties, information flow, non-interference, separation of duties; Software vulnerabilities: Memory corruption (stack-smashing, heap overflows, integer overflows), Input validation errors, Race conditions; Web server and Browser vulnerabilities; Malware and Untrusted software: Viruses and worms, Rootkits, Botnets, Obfuscation and evasion; Defenses for software threats: Static analysis for vulnerability detection, Code transformation for runtime policy checking, Runtime policy enforcement and sandboxing, Isolation and information-flow control; Virtual machines; Network-layer threats: network probing, scanning, Defenses - intrusion detection, Side-channel attacks - covert channels, timing attacks, power analysis, emanations, remanence and reuse; Human and social engineering - Digital rights management

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CS-436

COMPUTER VISION

Fundamentals of Computer Vision: Nature of Images, Homogeneous Transformations, Image Acquisition, Geometrical and Optical Image Formation, Human Encoding of Color, Color Spaces, Perspective Projection, Camera Technologies and Vision Systems Design; Basics of Image Processing: Filtering, Edge Detection, Features Detection, Contours, Segmentation, Morphological Operators; Motion: Detection, Optical Flow, Object Tracking, Motion Capture; Recognition: Large-Scale Instance Recognition and Retrieval, Category Recognition and Advanced Feature Encoding, Applications: OCRs, Facial Recognition, Quality Control, Visual Feedback, Mapping and Robot Guidance, Activity Monitoring, Motion Estimation, Autonomous Systems, Biomedical Imaging.

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CS-437

PARALLEL PROGRAMMING

Introduction of Parallel Programming; consideration of Parallel Programming; Classical applications; Models for Parallel Systems; Parallelization of Programs; Levels of Parallelism; Parallel Programming Patterns; Data Distributions; Parameterized Data Distribution; Information Exchange; Shared Variables; Communication Operations, concurrency and mutual exclusion; Parallel Algorithm design; parallel sorting and search; parallel matrix multiplication; parallel I/O; Performance Analysis of Parallel Programs; Benchmark; Performance

Metrics for Parallel Programs; Asymptotic Times for Global Communication; Message-Passing Programming; Interconnecting Topologies; Introduction to MPI; Collective Communication Operations; Process Groups and Communicators; Thread Programming; Creating and Merging Threads, Thread synchronization; thread scheduling; priority inversion; Introduction to Pthreads, Java Threads, OpenMP and CUDA; Introduction to Parallel Software Engineering and basic Patterns for Parallel Software Design.

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CS-406

COMPUTER ENGINEERING PROJECT

(0+6)

The final year students will be required to consult the Chairman of Computer Systems Engineering Department regarding the offering of various projects in the department. The student or group of students will be assigned the project by teacher concerned and will carry out the assignment as required and directed by the teacher. At the end of the academic session, they will submit the written report on work of their project to the chairman, preferably in the typed form. The students will be required to appear before a panel of examiners for oral examination.

By doing project the student will demonstrate the application of the knowledge gained during his/her stay in the university. He/she may also work on a project which may help him/her to learn more in some specific area related to Computer Engineering.

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